

Week 5- AWS

Assignment 19 – AWS Video Learning Challenge

Day-1 | Introduction to AWS Account | What is IAM ? | AWS Global Infrastructure



@Adrina_Colaco 8 minutes ago



Group 1 Adrina Colaco

3 key things I learned

- Difference between Root User and IAM User and how you can leverage these user accounts based on your need.
- Benefits of AWS global infrastructure to scale your application.
- Point of presence and Edge locations are same.

1 question or doubt you still have

- I would like to know a little more on the Cloud Front concept.

1 real-world scenario where I can apply it

- I would use CloudFront service for my website for faster delivery of content to the end users.

Show less



Reply

Day-2 | AWS S3 | Amazon S3 Buckets Deep Dive | Demo | AWS FREE COURSE



@Adrina_Colaco 0 seconds ago



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3 key things I learned

- The different storage categories offered by AWS.
- S3 bucket descriptions and Creation.
- S3 bucket policies, versioning, and encryption.

1 question or doubt I still have

- How is S3 preassigned URL significant and could you list some use cases?

1 real-world scenario where I can apply it

- I have already used S3 static website to launch a website for my small business.

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Day-3 | VPC explanation | AWS Virtual Private Cloud explained in 47 mins | 4K quality



@Adrina_Colaco 1 second ago



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3 key things I learned

- Use of VPC to create your private network.
- Networking concepts like Internet gateway, NAT Services.
- How to provide public access to a Private instance.

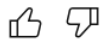
1 question or doubt you still have

- What is the difference between NAT Gateway and NAT Instance?

1 real-world scenario where I can apply it

- I would use VPC to create a private network for my 2 tier application.

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Day-4 | EC2 Deep Dive | AWS Elastic Compute Cloud | Hands-on Session | Live Session



@Adrina_Colaco 0 seconds ago



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3 Key Things I Learned

- How Amazon EC2 (Elastic Compute Cloud) works
- Fault tolerance, instance recovery, auto-scaling, health checks.
- How to distribute workload across multiple AZs.

1 Question or Doubt I Still Have

- How do you decide when to use spot/spot blocks vs reserved vs on-demand in a production system?

1 Real-World Scenario Where You Can Apply This

- To build a web application or microservice (for example, a SaaS product or internal tool for clients).

Show less



Day-5 | RDS Deep Dive | How to create AWS RDS | Amazon Relational Database | Project



@Adrina_Colaco 0 seconds ago



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3 Key Things I Learned

- Components & Architecture of AWS RDS
- How to Actually Create an RDS Instance Step-by-Step
- Things like choosing the right storage type (SSD vs magnetic), setting maintenance windows, enabling backups, and being careful with network/security settings (so your RDS is not exposed).

1 Question or Doubt I Still Have

- How do you handle schema migrations or major version upgrades in production without downtime (or with minimal downtime) when your database is large and in active use?

1 Real-World Scenario Where I Can Apply It

- Use RDS to build a SaaS application serving clients across India.
- Spin up an RDS instance using what you learned (selecting the proper instance class, enabling Multi-AZ for high availability, setting up read replicas for scaling reads).

Show less

Day 6 - Mastering AWS S3 Security: A Guide to Bucket Policies, Access Control & Identity Policies



@Adrina_Colaco 0 seconds ago



3 Key Things I Learned

- Least privilege in S3 policies is essential
- Use explicit Deny to override dangerous implicit Allows
- Condition keys and context-aware checks strengthen security

1 Question or Doubt I Still Have

- When combining multiple policies (bucket policy, IAM user policy, ACLs, etc.), how do you systematically debug or trace which policy is allowing or denying a particular S3 request? In other words: given a failed or allowed request, how can I pinpoint exactly which policy or condition caused it?

1 Real-World Scenario Where I Can Apply It

- Suppose your company is hosting user-uploaded files (images, documents) in an S3 bucket, and you build a web frontend and a mobile app to access them.

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Day-7 | What is AWS Cloudformation?



@Adrina_Colaco 0 seconds ago



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3 Key Things I Learned

- Template Structure & Sections
- Declarative vs Imperative Infrastructure
- Change Sets & Safe Updates

1 Question or Doubt I Still Have

- When your template needs to reference dynamically created resources (for example, a security group created in another stack, or an auto-generated name), how do you best manage dependencies and cross-stack references in a safe and maintainable way?

1 Real-World Scenario Where I Can Apply It

- As someone based in Pune (or anywhere), suppose you're setting up a new web application for your local business or side project. You can use CloudFormation to automate the provisioning:

Spin up your VPC, subnets, and security groups

Launch EC2 instances or Lambda functions

Create an RDS database

Set up S3 buckets, IAM roles, etc.

All-in-one template (or modular stacks)

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Day-8 | Deploy High Available Application with CloudFormation | 1 Demo Projects with Code



@Adrina_Colaco 0 seconds ago



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3 Key Things I Learned

- How Infrastructure as a code(IaC) helps to optimize deployment by saving time and improving accuracy.
- Creation of CloudFormation template
- Provisioning of resources using CloudFormation to deploy a complete application

1 Question or Doubt I Still Have

- Is there a way to automate the running of CloudFormation template across various environments like DEV, TEST, UAT instead of running the template manually in each of them?

1 Real-World Scenario Where I Can Apply It

- Provisioning EC2 for a Node.js app, RDS MySQL for the database, and S3 + CloudFront for React hosting

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Day -9: Deploy a Two-Tier Architecture Application on AWS



@Adrina_Colaco 0 seconds ago



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3 Key Things I Learned

- Understanding networking infrastructure and setting up of VPC, Subnets, Route Tables, Security Groups
- Setting up and configuration of RDS
- Mapping of Public IP to a Domain name using Route53.

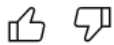
1 Question or Doubt I Still Have

- Is there a way to use Route53 if I have already purchased a domain name from GoDaddy?

1 Real-World Scenario Where I Can Apply It

- I would use this two-tier structure to set up my Recipe blog.

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Day - 10: Troubleshooting a Deployed AWS App Architecture - A Cloud Engineer's Guide



@Adrina_Colaco 1 second ago



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3 Key Things I Learned

- How to verify domain resolution, network connectivity, or whether DNS is pointing correctly
- Use of logs and monitoring for diagnostics
- Layered troubleshooting approach by isolating layers (DNS, networking, load balancing, backend, database) one by one, rather than attempting to fix everything at once.

1 Question / Doubt I Still Have

- When multiple layers like networking and application logs show anomalies simultaneously, what is the recommended strategy to prioritize which to fix first?

1 Real-World Scenario Where I Can Apply This

- Suppose you deploy your React front-end + Node.js backend + MySQL database on AWS via EC2, RDS, and load balancers. We can use this layered troubleshooting method to do root cause analysis when users report "site not reachable."

First check domain & DNS setup

Then check network / VPC / security groups

Then inspect load balancer health status

Then review instance logs (Node server)

Then database connectivity and health

Show less

Adrina

Day - 11 | AWS High Availability Architecture Deployment | Step-by-Step Guide | Part - (1/3)



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3 Key Things I Learned

- Design for Redundancy Across AZs to achieve high availability
- Use of Load Balancers and Auto Scaling to auto-provision.
- Failover and Multi-AZ Databases to enable automatic failover capability, so that the DB can switch to a standby in case of issues.

1 Question or Doubt I Still Have

- When using multi-AZ deployments for databases and auto scaling groups across AZs, how do you handle data consistency and latency between AZs (especially for writes)?

1 Real-World Scenario Where I Can Apply This

- To deploy the web servers in two AZs

Put an Elastic Load Balancer in front of them

Use Auto Scaling groups so if traffic surges (say a big launch), new instances spin up automatically

For the database (MySQL), enable Multi-AZ deployment (or cross-region replicas later)

That way, even if an AZ goes down (power or network issue), users won't see downtime.

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Day -12 | AWS High Availability Architecture Deployment | Step-by-Step Guide | Part - (2/3)



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3 Key Things I Learned

- Importance and Key aspects of achieving high availability in an application
- Automating the process of instance creation by using AMIs
- Set up Auto Scaling.

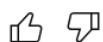
1 Question or Doubt I Still Have

- Can we delete the original EC2 instance used to create an AMI after we've set up the required processes, like auto scaling?

1 Real-World Scenario Where I Can Apply This

- To set up a highly available job application portal website or for a banking application.

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Day -13 | AWS High Availability Architecture Deployment | Step-by-Step Guide | Part - (3/3)



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3 Key Things I Learned

- Redundancy at the infrastructure and networking level.
- Creation of Read Replicas in RDS
- Point Route 53 to the DNS endpoint of the load balancer.

1 Question / Doubt I Still Have

- Why can't the read replica be created in the same AZ, and why do we need to enable Multi AZ?

1 Real-World Scenario Where I Can Apply It

- To build a full-stack web project that is highly available and accessible globally.

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Assignment 20- Deploy High Available Application with CloudFormation

1. Create the YAML script.



DevOpsHigh
AvailableApp_CloudF

2. Create a Stack in CloudFormation.

Step 1

Create stack

Step 2

Specify stack details

Step 3

Configure stack options

Step 4

Review and create

Create stack

Prerequisite - Prepare template

You can also create a template by scanning your existing resources in the [IaC generator](#).

Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Choose an existing template

Upload or choose an existing template.

☐ Build from Infrastructure Composer

Create a template using a visual builder.

Specify template

[Info](#)

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

Template source

Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

☐ Amazon S3 URL

Provide an Amazon S3 URL to your template.

☒ Upload a template file

Upload your template directly to the console.

☐ Sync from Git

Sync a template from your Git repository.

Upload a template file

Choose file

DevOpsHigh AvailableApp_CloudFormation.yaml

JSON or YAML formatted file

S3 URL: https://s3.us-east-1.amazonaws.com/cf-templates-deh83tgw5a8-us-east-1/2025-09-29T100314.547Zcuy-DevOpsHighAvailableApp_CloudFormation.yaml

[View in Infrastructure Composer](#)

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CloudFormation

Stacks

Create stack

Account ID: 3111-4154-2113

Adriana Colaco

Delete initiated for am:aws.cloudformation:us-east-1:311141542113:stack/DevOpsHAApplication/e39dcf90-9d20-11f0-b4a5-0e25984a9e07

Step 1

Create stack

Step 2

Specify stack details

Step 3

Configure stack options

Step 4

Review and create

Specify stack details

Provide a stack name

Stack name

DevOpsHAApplication

Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 19/128.

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

AmazonLinuxAMIId

/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2

HAwebVpcCidr

10.0.0.0/20

InstanceType

t3.micro

KeyName

EC2 Key Pair for SSH

DevopsAppHosting

PrivateSubnet1Cidr

10.0.2.0/24

PrivateSubnet2Cidr

10.0.12.0/24

PublicSubnet1Cidr

10.0.1.0/24

PublicSubnet2Cidr

10.0.11.0/24

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CloudFormation > Stacks > Create stack

Step 1
Create stack

Step 2
Specify stack details

Step 3
Configure stack options

Step 4
Review and create

Configure stack options

Tags - optional

Tags (key-value pairs) are used to apply metadata to AWS resources, which can help in organizing, identifying, and categorizing those resources. You can add up to 50 unique tags for each stack.

No tags associated with the stack.

Add new tag

You can add 50 more tag(s)

Permissions - optional

Specify an existing AWS Identity and Access Management (IAM) service role that CloudFormation can assume.

IAM role - optional

Choose the IAM role for CloudFormation to use for all operations performed on the stack.

IAM role name

Sample-role-name

Remove

Stack failure options

Behavior on provisioning failure

Specify the roll back behavior for a stack failure. [Learn more](#)

Roll back all stack resources

Roll back the stack to the last known stable state.

Preserve successfully provisioned resources

Preserves the state of successfully provisioned resources, while rolling back failed resources to the last known stable state. Resources without a last known stable state will be deleted upon the next stack operation.

Delete newly created resources during a rollback

Specify whether resources that were created during a failed operation should be deleted regardless of their deletion policy. [Learn more](#)

Use deletion policy

Retains or deletes created resources according to their attached deletion policy.

Delete all newly created resources

Deletes created resources during a rollback regardless of their attached deletion policy.

Additional settings

You can set additional options for your stack, like notification options and a stack policy. [Learn more](#)

Stack policy - optional

Defines the resources that you want to protect from unintentional updates during a stack update.

Rollback configuration - optional

Specify alarms for CloudFormation to monitor when creating and updating the stack. If the operation breaches an alarm threshold, CloudFormation rolls it back.

Notification options - optional

Specify a new or existing Amazon Simple Notification Service topic where notifications about stack events are sent.

Stack creation options - optional

Specify the timeout and termination protection options for stack creation.

Cancel Previous Next

CloudFormation > Stacks > DevOpsHAAApplication

Filter status

Active

View nested

Stacks

DevOpsHAAApplication

2025-09-29 16:11:43 UTC+0530

CREATE_IN_PROGRESS

Stack info Events Resources Outputs Parameters Template Change sets Git sync

Table view Timeline view

Events (1)

Search events

Timestamp	Logical ID	Status	Detailed status	Status reason	Hook invocations
2025-09-29 16:11:43 UTC+0530	DevOpsHAAApplication	CREATE_IN_PROGRESS	-	User Initiated	-

Stacks (1)

Filter status

Active

View nested

Stacks

DevOpsHAAApplication

2025-09-29 16:31:36 UTC+0530

CREATE_COMPLETE

DevOpsHAAApplication

Delete

Update stack

Stack actions

Create stack

Stack info

Events

Resources

Outputs

Parameters

Template

Change sets

Git sync

Table view

Timeline view

Events (87)

Search events

Timestamp	Logical ID	Status	Detailed status	Status reason
2025-09-29 16:35:13 UTC+0530	DevOpsHAAApplication	CREATE_COMPLETE	-	-
2025-09-29 16:35:11 UTC+0530	ALBListener	CREATE_COMPLETE	-	-
2025-09-29 16:35:10 UTC+0530	ALBListener	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:35:09 UTC+0530	ALBListener	CREATE_IN_PROGRESS	-	-
2025-09-29 16:35:08 UTC+0530	ALB	CREATE_COMPLETE	-	-
2025-09-29 16:33:40 UTC+0530	PrivateRoute1	CREATE_COMPLETE	-	-
2025-09-29 16:33:40 UTC+0530	PublicRoute2	CREATE_COMPLETE	-	-
2025-09-29 16:33:39 UTC+0530	PrivateRoute1	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:33:39 UTC+0530	PublicRoute2	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:33:38 UTC+0530	PrivateRoute1	CREATE_IN_PROGRESS	-	-
2025-09-29 16:33:38 UTC+0530	PublicRoute2	CREATE_IN_PROGRESS	-	-
2025-09-29 16:33:37 UTC+0530	NatGateway	CREATE_COMPLETE	-	-
2025-09-29 16:32:52 UTC+0530	AutoScalingGroup	CREATE_COMPLETE	-	-
2025-09-29 16:32:26 UTC+0530	AutoScalingGroup	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE	Eventual consistency check initiated
2025-09-29 16:32:16 UTC+0530	NatEIP	CREATE_COMPLETE	-	-
2025-09-29 16:32:16 UTC+0530	AutoScalingGroup	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:32:15 UTC+0530	AutoScalingGroup	CREATE_IN_PROGRESS	-	-
2025-09-29 16:32:14 UTC+0530	LaunchTemplate	CREATE_COMPLETE	-	-
2025-09-29 16:32:13 UTC+0530	LaunchTemplate	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:32:12 UTC+0530	PublicRoute	CREATE_COMPLETE	-	-
2025-09-29 16:32:12 UTC+0530	LaunchTemplate	CREATE_IN_PROGRESS	-	-
2025-09-29 16:32:11 UTC+0530	EC2SecurityGroup	CREATE_COMPLETE	-	-
2025-09-29 16:32:09 UTC+0530	ALBTargetGroup	CREATE_COMPLETE	-	-
2025-09-29 16:32:07 UTC+0530	PublicRouteTable1Association	CREATE_COMPLETE	-	-
2025-09-29 16:32:07 UTC+0530	PrivateRouteTableAssociation2	CREATE_COMPLETE	-	-
2025-09-29 16:32:07 UTC+0530	PrivateRouteTableAssociation1	CREATE_COMPLETE	-	-
2025-09-29 16:32:07 UTC+0530	PublicRouteTable2Association	CREATE_COMPLETE	-	-
2025-09-29 16:32:07 UTC+0530	PublicRoute	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE	Eventual consistency check initiated
2025-09-29 16:32:07 UTC+0530	PrivateRouteTableAssociation2	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:32:05 UTC+0530	PrivateRouteTable2	CREATE_COMPLETE	-	-
2025-09-29 16:32:05 UTC+0530	PublicRouteTable	CREATE_COMPLETE	-	-
2025-09-29 16:32:05 UTC+0530	EC2SecurityGroup	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:32:04 UTC+0530	PrivateRouteTable1	CREATE_COMPLETE	-	-
2025-09-29 16:32:02 UTC+0530	ALBSecurityGroup	CREATE_COMPLETE	-	-
2025-09-29 16:32:02 UTC+0530	NatGateway	CREATE_IN_PROGRESS	-	-
2025-09-29 16:32:01 UTC+0530	NatEIP	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE	Eventual consistency check initiated
2025-09-29 16:32:00 UTC+0530	PublicSubnet2	CREATE_COMPLETE	-	-
2025-09-29 16:31:59 UTC+0530	VPCtoIGWConnection	CREATE_COMPLETE	-	-
2025-09-29 16:31:57 UTC+0530	PrivateSubnet1	CREATE_COMPLETE	-	-
2025-09-29 16:31:57 UTC+0530	PrivateSubnet2	CREATE_COMPLETE	-	-
2025-09-29 16:31:57 UTC+0530	VPCtoIGWConnection	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:31:56 UTC+0530	PublicSubnet1	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:31:55 UTC+0530	IGW	CREATE_COMPLETE	-	-
2025-09-29 16:31:55 UTC+0530	ALBSecurityGroup	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:31:55 UTC+0530	PrivateRouteTable2	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE	Eventual consistency check initiated
2025-09-29 16:31:54 UTC+0530	PrivateSubnet1	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:31:54 UTC+0530	PrivateRouteTable2	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:31:52 UTC+0530	HAWebVPC	CREATE_COMPLETE	-	-
2025-09-29 16:31:40 UTC+0530	IGW	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE	Eventual consistency check initiated
2025-09-29 16:31:40 UTC+0530	HAWebVPC	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:31:40 UTC+0530	IGW	CREATE_IN_PROGRESS	-	Resource creation initiated
2025-09-29 16:31:39 UTC+0530	IGW	CREATE_IN_PROGRESS	-	-
2025-09-29 16:31:39 UTC+0530	HAWebVPC	CREATE_IN_PROGRESS	-	-

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3. Verify if resources are created

a. VPC

Your VPCs (1/2) [Info](#)

Find VPCs by attribute or tag

Last updated 1 minute ago [Actions](#) [Create VPC](#)

☐	Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
☐	DoNotDelete	vpc-00d70ddc6828a5629	Available	Off	172.31.0.0/16	-	dopt-00b929515
☒	HAweb VPC	vpc-026f156a1afd0f345	Available	Off	10.0.0.0/20	-	dopt-00b929515

vpc-026f156a1afd0f345 / HAweb VPC

[Details](#) [Resource map](#) [CIDRs](#) [Flow logs](#) [Tags](#) [Integrations](#)

Details

VPC ID
[vpc-026f156a1afd0f345](#)

DNS resolution
Enabled

Main network ACL
[acl-03f993ddcd70bd9df](#)

IPv6 CIDR (Network border group)
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Block Public Access
Off

DHCP option set
[dopt-00b92951508cb2a3b](#)

IPv4 CIDR
10.0.0.0/20

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Enabled

Main route table
[rtb-0c3fe0fc18033ba30](#)

IPv6 pool
-

Owner ID
[311141542113](#)

b. Internet Gateway

Internet gateways (1/2) [Info](#)

Find Internet gateways by attribute or tag

[Actions](#) [Create internet gateway](#)

☐	Name	Internet gateway ID	State	VPC ID	Owner
☒	HAweb IGW	igw-0fa4b4acf4a51074c	Attached	vpc-026f156a1afd0f345 HAweb VPC	311141542113
☐	-	igw-0fe1e9884fd2a57f8	Attached	vpc-00d70ddc6828a5629 DoNotDelete	311141542113

igw-0fa4b4acf4a51074c / HAweb IGW

[Details](#) [Tags](#)

Details

Internet gateway ID
[igw-0fa4b4acf4a51074c](#)

State
Attached

VPC ID
[vpc-026f156a1afd0f345 | HAweb VPC](#)

Owner
[311141542113](#)

c. Public and Private Subnets

Subnets (4/15) Info

Last updated 3 minutes ago

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
Public Subnet 2	subnet-05f62f92bf3081bc2	Available	vpc-026f156a1afd0f345 HAweb VPC	Off	10.0.11.0/24
Public Subnet 1	subnet-0b314c135df030b6e	Available	vpc-026f156a1afd0f345 HAweb VPC	Off	10.0.1.0/24
Private Subnet 2	subnet-0a2189624a5809168	Available	vpc-026f156a1afd0f345 HAweb VPC	Off	10.0.12.0/24
Private Subnet 1	subnet-070d875d7d2d79223	Available	vpc-026f156a1afd0f345 HAweb VPC	Off	10.0.2.0/24

Subnets: subnet-070d875d7d2d79223, subnet-0a2189624a5809168, subnet-0b314c135df030b6e, subnet-05f62f92bf3081bc2

d. Public and Private Route Tables

Route tables (3/6) Info

Last updated 4 minutes ago

Find route tables by attribute or tag

Clear filters

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Own
Private Route Table 1	rtb-0c5711311c04ff27e	subnet-070d875d7d2d79...	-	No	vpc-026f156a1afd0f345 HAweb VPC	3111
Private Route Table 2	rtb-018c89874e351a508	subnet-0a2189624a5809...	-	No	vpc-026f156a1afd0f345 HAweb VPC	3111
Public Route Table	rtb-022cb03de2732d143	2 subnets	-	No	vpc-026f156a1afd0f345 HAweb VPC	3111

Route tables: rtb-0c5711311c04ff27e, rtb-018c89874e351a508, rtb-022cb03de2732d143

e. Security Groups

sg-0b08c20b2ef65c5cd - DevOpsHAApplication-EC2SecurityGroup-vNCvyaBD6Zlp

Actions

Details

Security group name DevOpsHAApplication-EC2SecurityGroup-vNCvyaBD6Zlp	Security group ID sg-0b08c20b2ef65c5cd	Description Allow traffic from ALB	VPC ID vpc-026f156a1afd0f345
Owner 311141542113	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sg-0a24a21807319acd3	-	HTTP	TCP	80	sg-0dca6c57e4892506...

sg-0dca6c57e48925060 - App

Actions

Details

Security group name App	Security group ID sg-0dca6c57e48925060	Description Enable access to App	VPC ID vpc-026f156a1afd0f345
Owner 311141542113	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sg-0cfb259263bb029bb	IPv4	HTTP	TCP	80	0.0.0.0/0
-	sg-0ac6d3243b62fa30f	IPv4	SSH	TCP	22	0.0.0.0/0

f. Load Balancer

Load balancers (1/1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers										< 1 >	
☒	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups			
☒	HA-ALB	Active	application	Internet-facing	IPv4	vpc-026f156a1afd0f345	2 Availability Zones	sg-0dca6c57e489			

Load balancer: HA-ALB

Details

Load balancer type
Application

Status
Active

VPC
[vpc-026f156a1afd0f345](#)

Load balancer IP address type
IPv4

Scheme
Internet-facing

Hosted zone
Z35SXDOTRQ7X7K

Availability Zones
[subnet-0b314c135df030b6e](#) us-east-1a (use1-az2)
[subnet-05f62f92bf3081bc2](#) us-east-1b (use1-az4)

Date created
September 29, 2025, 16:32 (UTC+05:30)

Load balancer ARN
[arn:aws:elasticloadbalancing:us-east-1:311141542113:loadbalancer/app/HA-ALB/a653f5169ed6365c](#)

DNS name info
[HA-ALB-1109374729.us-east-1.elb.amazonaws.com](#) (A Record)

EC2 > Load balancers > HA-ALB

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

CloudShell

Feedback

HA-ALB

▼ Details

Load balancer type

Application

Status

Active

Scheme

Internet-facing

VPC

[vpc-026f156a1afd0f345](#)

Availability Zones

[subnet-0b314c135df030b6e](#) us-east-1a (use1-az2)
[subnet-05f62f92bf3081bc2](#) us-east-1b (use1-az4)

Load balancer IP address type

IPv4

Date created

September 29, 2025, 16:32 (UTC+05:30)

Load balancer ARN

[arn:aws:elasticloadbalancing:us-east-1:311141542113:loadbalancer/app/HA-ALB/a653f5169ed6365c](#)

DNS name info

[HA-ALB-1109374729.us-east-1.elb.amazonaws.com](#) (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1) Info

Filter listeners

Manage rules

Manage listener

Add listener

g. Target Group & Health Checks

DevOps-ALBta-WR6IVEUTZTVE

Details

arn:aws:elasticloadbalancing:us-east-1:311141542113:targetgroup/DevOps-ALBta-WR6IVEUTZTVE/9116b89f60a8f4eb

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-026f156a1afd0f345
IP address type IPv4	Load balancer HA-ALB		

2 Total targets	2 Healthy	0 Unhealthy	0 Unused	0 Initial	0 Draining
--------------------	--------------	----------------	-------------	--------------	---------------

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (2) [Info](#)

Anomaly mitigation: **Not applicable** [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

h. Launch Template

LaunchTemplate_mirhYPreXhse (lt-0ba79825fa0292abf)

Launch template details

Launch template ID lt-0ba79825fa0292abf	Launch template name LaunchTemplate_mirhYPreXhse	Default version 1	Owner arn:aws:iam::311141542113:root
--	---	----------------------	---

Details | Versions | Template tags

Launch template version details

Version 1 (Default)	Description -	Date created 2025-09-29T11:02:13.000Z	Created by arn:aws:iam::311141542113:root
------------------------	------------------	--	--

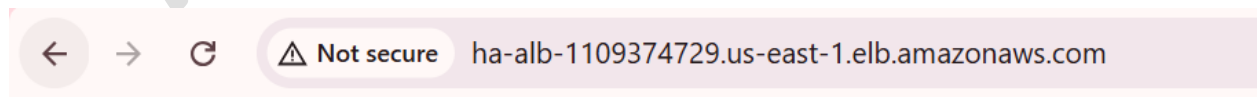
Instance details | Storage | Resource tags | Network interfaces | Advanced details

AMI ID ami-0254b2d5c4c472488	Instance type t3.micro	Availability Zone -	Availability Zone Id -
Key pair name DevopsAppHosting	Security groups -	Security group IDs sg-0b08c20b2ef65c5cd	

i. Auto scaling Group

The screenshot displays the AWS Management Console interface for an Auto Scaling group. The left sidebar shows the navigation menu with categories like Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area shows the details for the Auto Scaling group 'DevOpsHAAApplication-AutoScalingGroup-uaCSQ084G99s'. The 'Capacity overview' section shows a desired capacity of 2 instances, scaling limits from 2 to 4, and a status of 'In Service'. The 'Launch template' section shows the use of launch template 'lt-0ba79825fa0292abf' with AMI ID 'ami-0254b2d5c4c472488' and instance type 't3.micro'. The 'Health checks' section shows the group is 'Not protected from scale in' and has a default termination policy. The 'Tags' section shows a system tag 'aws:cloudformationstack-name' with the value 'DevOpsHAAApplication'.

4. Verify your application is accessible via the Load Balancer DNS name.



**Hello from EC2 in Auto Scaling Group.
Deployed By Adrina Colaco.**

5. Stop/terminate one EC2 instance and confirm the Load Balancer routes traffic to the remaining healthy instance.

The screenshot shows the AWS Management Console interface. At the top, a green banner indicates 'Successfully initiated stopping of i-09c35f5f7dc4058c8'. Below this, the 'Instances (1/2)' table lists two instances. The first instance, i-0690849c78760795e, is in a 'Running' state. The second instance, i-09c35f5f7dc4058c8, is in a 'Stopped' state. The details for the stopped instance are expanded, showing its instance ID, IP addresses, and state. Below the console, a browser address bar shows the URL 'ha-alb-1109374729.us-east-1.elb.amazonaws.com' with a 'Not secure' warning.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
	i-0690849c78760795e	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-18-207-228-241.co...	18.207.2
	i-09c35f5f7dc4058c8	Stopped	t3.micro	-	View alarms +	us-east-1b	-	-

i-09c35f5f7dc4058c8

Instance summary

Instance ID: i-09c35f5f7dc4058c8

IPV6 address: -

Hostname type: IP name: ip-10-0-12-180.ec2.internal

Answer private resource DNS name

Public IPv4 address: -

Instance state: Stopped

Private IP DNS name (IPv4 only): ip-10-0-12-180.ec2.internal

Instance type

Private IPv4 addresses: 10.0.12.180

Public DNS: -

Elastic IP addresses

Browser address bar: Not secure ha-alb-1109374729.us-east-1.elb.amazonaws.com

**Hello from EC2 in Auto Scaling Group.
Deployed By Adrina Colaco.**


HA_App_CloudFormation.mp4

Assignment 21- EpicBooks on AWS with CloudFormation (optional)

1. Create YAML Script.


EpicBooksApp_CloudFormation.yaml

2. Create CloudFormation Stack.

3. Verify Resources.

- a. VPC,
- b. one **public subnet** (EC2 web)
- c. **two private subnets** (RDS MySQL)
- d. SGs
- e. EC2 with **UserData** (nginx placeholder page)
- f. **RDS**



EpicBook_CloudFormation.mp4

4. Verify URL.

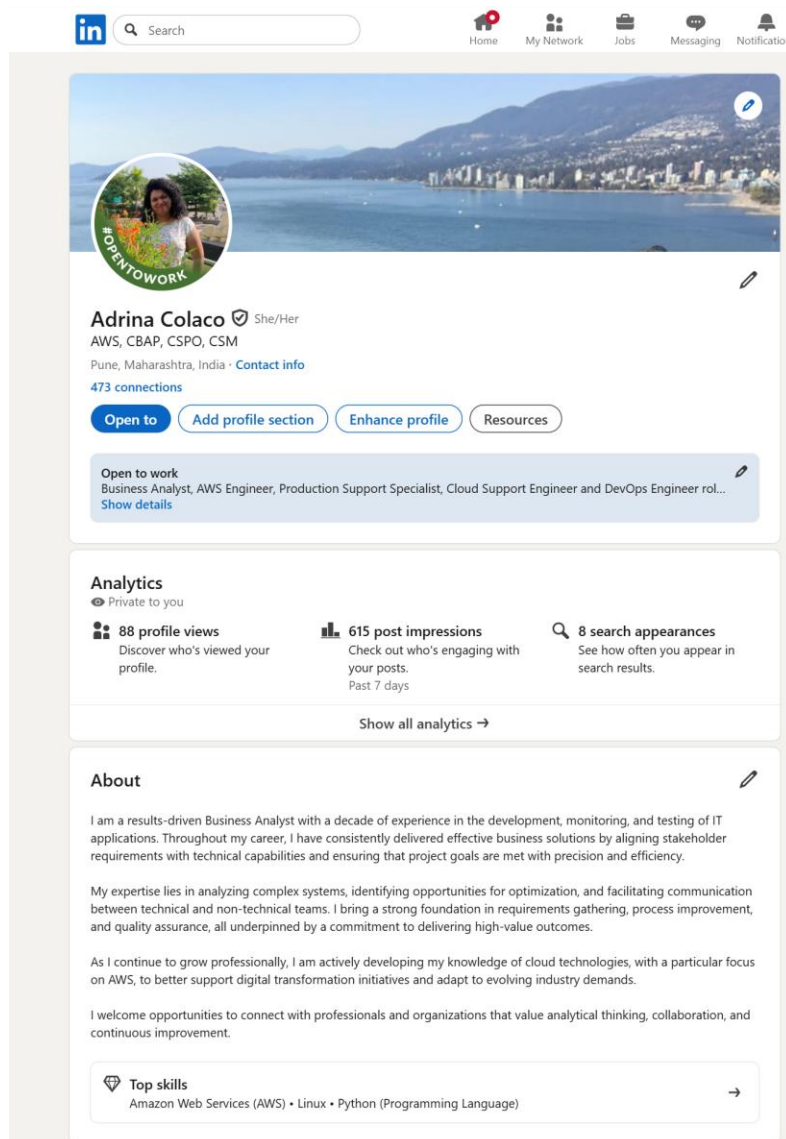
The screenshot shows the AWS CloudFormation console. On the left, the 'Stacks (1)' sidebar lists the 'EpicBooksCloudFormation' stack with a status of 'CREATE_COMPLETE'. The main panel shows the 'Outputs (1)' tab for this stack. A table lists the output 'WebsiteURL' with the value 'http://ec2-54-152-210-63.compute-1.amazonaws.com'. A red arrow points to this URL. The description for this output is 'Access the EpicBook app via EC2 Public DNS'.

Key	Value	Description	Export name
WebsiteURL	http://ec2-54-152-210-63.compute-1.amazonaws.com	Access the EpicBook app via EC2 Public DNS	-

Assignment 22 – LinkedIn Profile Optimization- Do & Review

Part A - Optimize Your Profile

Before



The screenshot shows a LinkedIn profile for Adrina Colaco. The header includes the LinkedIn logo, a search bar, and navigation icons for Home, My Network, Jobs, Messaging, and Notifications. The profile banner features a scenic view of a city by the water. The profile picture is a circular headshot of Adrina Colaco, with a green 'OPEN TO WORK' badge overlaid. Below the profile picture, the name 'Adrina Colaco' is displayed with a verified badge and the pronouns 'She/Her'. The current position is 'AWS, CBAP, CSPO, CSM' in 'Pune, Maharashtra, India'. There are 473 connections. Action buttons include 'Open to', 'Add profile section', 'Enhance profile', and 'Resources'. A section titled 'Open to work' lists roles: Business Analyst, AWS Engineer, Production Support Specialist, Cloud Support Engineer, and DevOps Engineer. Below this is an 'Analytics' section showing 88 profile views, 615 post impressions, and 8 search appearances. The 'About' section contains a detailed bio. At the bottom, a 'Top skills' section lists 'Amazon Web Services (AWS)', 'Linux', and 'Python (Programming Language)'.

Adrina Colaco ✓ She/Her
AWS, CBAP, CSPO, CSM
Pune, Maharashtra, India • [Contact info](#)
473 connections

[Open to](#) [Add profile section](#) [Enhance profile](#) [Resources](#)

Open to work
Business Analyst, AWS Engineer, Production Support Specialist, Cloud Support Engineer and DevOps Engineer rol...
[Show details](#)

Analytics
Private to you

- 88 profile views**
Discover who's viewed your profile.
- 615 post impressions**
Check out who's engaging with your posts.
Past 7 days
- 8 search appearances**
See how often you appear in search results.

[Show all analytics →](#)

About

I am a results-driven Business Analyst with a decade of experience in the development, monitoring, and testing of IT applications. Throughout my career, I have consistently delivered effective business solutions by aligning stakeholder requirements with technical capabilities and ensuring that project goals are met with precision and efficiency.

My expertise lies in analyzing complex systems, identifying opportunities for optimization, and facilitating communication between technical and non-technical teams. I bring a strong foundation in requirements gathering, process improvement, and quality assurance, all underpinned by a commitment to delivering high-value outcomes.

As I continue to grow professionally, I am actively developing my knowledge of cloud technologies, with a particular focus on AWS, to better support digital transformation initiatives and adapt to evolving industry demands.

I welcome opportunities to connect with professionals and organizations that value analytical thinking, collaboration, and continuous improvement.

Top skills
Amazon Web Services (AWS) • Linux • Python (Programming Language)

After

Business Analyst | Cloud & ERP Solutions | Agile Delivery | AWS Certified






Adrina Colaco She/Her
Business Analyst | Cloud & ERP Solutions | AWS Certified | Agile Delivery & Production Support | Open to New Opportunities
Pune, Maharashtra, India · [Contact info](#)
473 connections
[Open to](#) [Add profile section](#) [Resources](#)

Open to work
Business Analyst, AWS Engineer, Production Support Specialist, Cloud Support Engineer and DevOps Engineer rol...
[Show details](#)

Analytics
Private to you

 **88 profile views**
Discover who's viewed your profile.

 **566 post impressions**
Check out who's engaging with your posts.
Past 7 days

 **8 search appearances**
See how often you appear in search results.

[Show all analytics →](#)

About

I am a dynamic Business Analyst and Cloud Solutions professional with 10+ years of experience spanning ERP systems, AWS cloud projects, production support, and Agile delivery. My career journey bridges business and technology; from leading requirement gathering and ERP enhancements in enterprise systems, to building and migrating modern cloud solutions with AWS.

Key Highlights

AWS Cloud Projects: Built secure web applications, migrated on-premise servers to AWS, automated deployments with Docker pipelines, and monitored infrastructure with CloudWatch, CloudTrail & SNS.

ERP & Business Analysis: Delivered HCM module enhancements, automated processes saving >100 hours/month, and developed interactive HR dashboards in Tableau.

Global IT Delivery: Led cross-functional teams at TCS and the State of Tennessee, creating business-critical applications across finance, insurance, and manufacturing domains.

Entrepreneurial Mindset: Founded and scaled an eco-friendly e-commerce venture, managing end-to-end operations and digital presence.

Certifications: AWS Certified Cloud Practitioner, CBAP (Business Analysis), Scrum Master & Product Owner (CSM, CSPO), ITIL, Lean Six Sigma, SnowPro Associate.

What I bring: A blend of technical expertise, stakeholder collaboration, and process optimization, with proven ability to turn complex business needs into scalable, cloud-ready technology solutions.

Currently exploring opportunities in Business Analysis, Cloud/ERP solutions, and Agile delivery roles; where I can leverage my background to drive innovation, efficiency, and impact.

Top skills
Amazon Web Services (AWS) • Linux • Python (Programming Language)

Part B – Peer Review

Not possible to do this activity as LinkedIn profile cannot be used in YouTube comments.

LinkedIn Post_1:

https://www.linkedin.com/posts/adrina-colaco_unlocking-the-power-of-aws-infrastructure-activity-7378429376843137024-61NB?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

[61NB?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/posts/adrina-colaco_unlocking-the-power-of-aws-infrastructure-activity-7378429376843137024-61NB?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

LinkedIn Post_2:

https://www.linkedin.com/posts/adrina-colaco_cloudcomputing-aws-cloudformation-activity-7378432334620037121-oj2C?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

[oj2C?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/posts/adrina-colaco_cloudcomputing-aws-cloudformation-activity-7378432334620037121-oj2C?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

Blog Post:

<https://medium.com/@adcola13/unlocking-the-power-of-aws-infrastructure-composer-9545767ce5e5>

The End

P.S. This document is part of the **FREE DevOps Micro Internship (DMI) Cohort** run by [Pravin Mishra](#). You can start your DevOps journey for free from his [YouTube Playlist](#).